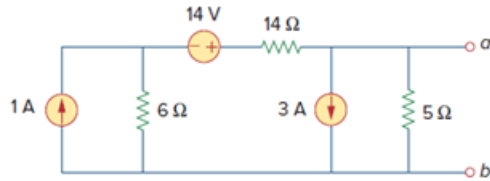


Problem

Find the Thevenin and Norton equivalents at terminals a - b of the circuit shown in Fig. 4.108.

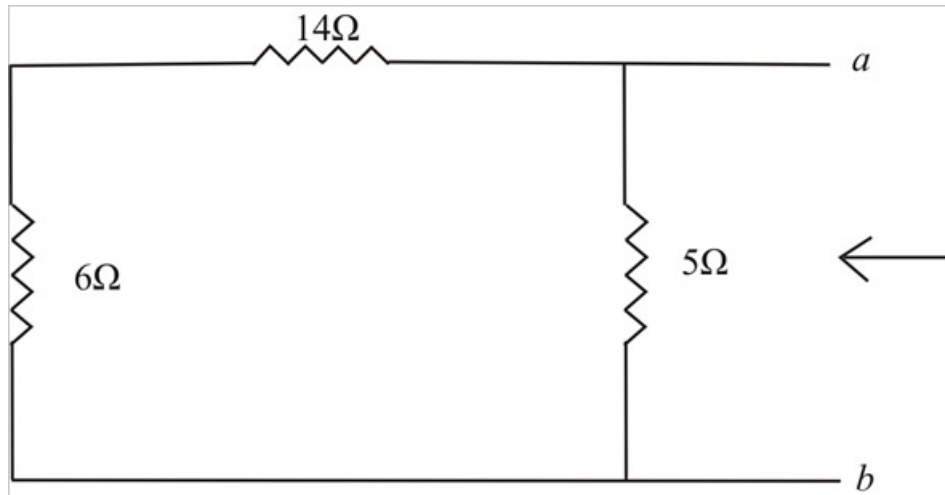
Figure. 4.108:



Step-by-step solution

Step 1 of 6

To find R_{Th} , Consider the circuit below.



Step 2 of 6

$$R_{Th} = (6\Omega + 14\Omega) \parallel 5\Omega$$

$$R_{Th} = \frac{5 \times 20}{5 + 20}$$

$$R_{Th} = \frac{100}{25}$$

$$R_{Th} = 4\Omega$$

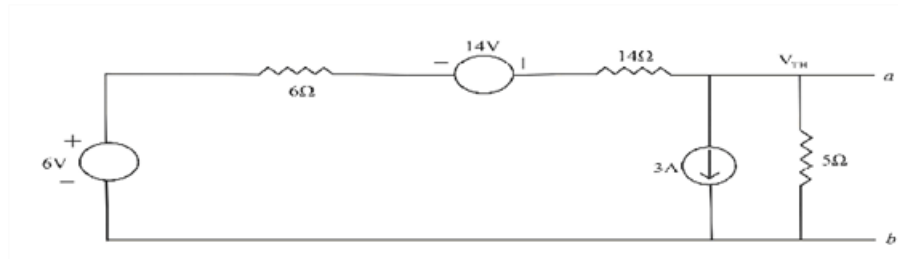
Therefore,

$$R_{Th} = R_N$$

$$R_N = 4\Omega$$

Step 3 of 6

Applying source transformation to the 1A current source, we obtain the circuit below.



Step 4 of 6

At node a,

$$\frac{14\text{ V} + 6\text{ V} - V_{Th}}{6\Omega + 14\Omega} = 3 + \frac{V_{Th}}{5}$$

$$\frac{20 - V_{Th}}{20} = \frac{15 + V_{Th}}{5}$$

$$20 - V_{Th} = 60 + 4V_{Th}$$

$$5V_{Th} = -40$$

$$V_{Th} = -8\text{ V}$$

Therefore,

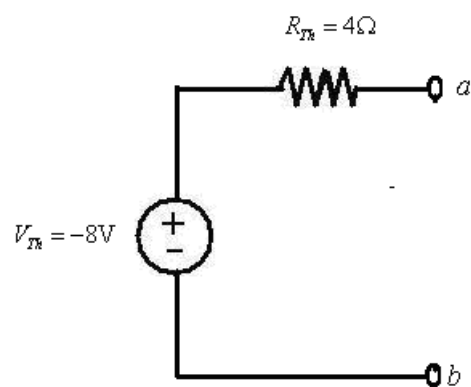
$$I_N = \frac{V_{Th}}{R_{Th}}$$

$$I_N = \frac{-8}{4}$$

$$I_N = -2\text{ A}$$

Step 5 of 6

Therefore, Thevenin's equivalent circuit as shown in below;



Step 6 of 6

Therefore, the Norton equivalent circuit as shown in below

